

Practical No:02

Input:-

```
CREATE DATABASE IF NOT EXISTS StudentDB;  
USE StudentDB;
```

```
DROP TABLE IF EXISTS Enrollment;  
DROP TABLE IF EXISTS Students;  
DROP TABLE IF EXISTS Courses;
```

```
CREATE TABLE Students (  
    student_id INT PRIMARY KEY,  
    name VARCHAR(50) NOT NULL,  
    age INT,  
    department VARCHAR(50)  
);
```

```
CREATE TABLE Courses (  
    course_id INT PRIMARY KEY,  
    course_name VARCHAR(50) NOT NULL,  
    credits INT  
);
```

```
CREATE TABLE Enrollment (  
    student_id INT,  
    course_id INT,  
    grade CHAR(1),  
    PRIMARY KEY (student_id, course_id),  
    FOREIGN KEY (student_id) REFERENCES Students(student_id)  
        ON DELETE CASCADE  
        ON UPDATE CASCADE,  
    FOREIGN KEY (course_id) REFERENCES Courses(course_id)  
        ON DELETE CASCADE  
        ON UPDATE CASCADE  
);
```

```
INSERT INTO Students VALUES  
(1,'Alice',20,'CS'),  
(2,'Bob',21,'EE'),  
(3,'Charlie',22,'ME'),  
(4,'David',23,'CS');
```

```
INSERT INTO Courses VALUES  
(101,'DBMS',4),  
(102,'OS',3),  
(103,'Networks',3),  
(104,'AI',4);
```

```
INSERT INTO Enrollment VALUES  
(1,101,'A'),  
(1,102,'B'),  
(2,101,'B'),  
(3,103,'A'),  
(4,101,'B');
```

```
(4,104,'A');
```

```
SELECT * FROM Students WHERE age > 21;
```

```
SELECT * FROM Students WHERE department = 'CS';
```

```
SELECT * FROM Students ORDER BY name ASC;
```

```
SELECT * FROM Courses ORDER BY credits DESC;
```

```
SELECT c.course_name, COUNT(e.student_id) AS student_count  
FROM Courses c  
JOIN Enrollment e ON c.course_id = e.course_id  
GROUP BY c.course_name;
```

```
SELECT department, AVG(age) AS avg_age  
FROM Students  
GROUP BY department;
```

```
SELECT name  
FROM Students  
WHERE student_id IN (  
    SELECT student_id  
    FROM Enrollment  
    WHERE course_id IN (  
        SELECT course_id  
        FROM Courses  
        WHERE credits > 3  
    )  
);
```

```
SELECT course_name  
FROM Courses  
WHERE course_id IN (  
    SELECT course_id  
    FROM Enrollment  
    GROUP BY course_id  
    HAVING COUNT(student_id) > 1  
);
```

```
CREATE INDEX idx_student_age ON Students(age);  
CREATE INDEX idx_course_credits ON Courses(credits);  
CREATE INDEX idx_enrollment ON Enrollment(student_id, course_id);
```

```
EXPLAIN SELECT * FROM Students WHERE age > 21;  
EXPLAIN SELECT * FROM Courses WHERE credits > 3;
```

```
SELECT * FROM Students;  
SELECT * FROM Courses;  
SELECT * FROM Enrollment;  
SELECT  
    s.student_id,  
    s.name,  
    s.department,  
    c.course_name,
```

```
e.grade  
FROM Students s  
JOIN Enrollment e ON s.student_id = e.student_id  
JOIN Courses c ON e.course_id = c.course_id;
```

Output:-

Result Grid   Filter Rows: Export:  Wrap Cell					
	student_id	name	department	course_name	grade
▶	1	Alice	CS	DBMS	A
	1	Alice	CS	OS	B
	2	Bob	EE	DBMS	B
	3	Charlie	ME	Networks	A
	4	David	CS	DBMS	B
	4	David	CS	AI	A